

2.2 Matrices

Further Pure 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A Level matrix work extends to three-dimensional transformations, eigenvalues and diagonalisation where applicable. Solve $\det(A - \lambda I) = 0$ for eigenvalues; an eigenvector satisfies $(A - \lambda I)v = 0$. Repeated eigenvalues need enough independent eigenvectors for diagonalisation.

Worked example: follow the reasoning

For $A = \text{diag}(2, 3, 4)$, characteristic equation is $(2 - \lambda)(3 - \lambda)(4 - \lambda) = 0$, so eigenvalues are 2, 3, 4.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 For $A = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$, give eigenvalues and one eigenvector for each.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Eigenvalues 3 and 5 [1]; $v_3=(1,0)$ [1], $v_5=(0,1)$ [1]; nonzero scalar multiples accepted.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.